

Task 2 – Student Profile Using JavaScript

Aim

To create a Student Profile application using JavaScript classes, objects, DOM manipulation, and event handling. The application accepts student details from the user and dynamically displays the student profile on the webpage when the button is clicked.

Description

This application allows the user to enter student details such as Name, Roll Number, Department, and CGPA. A JavaScript Student class is created with these properties. When the Create Profile button is clicked, the values entered by the user are retrieved from the input fields and used to create an object of the Student class.

Using DOM manipulation, HTML elements are dynamically created to display the student's details. The generated elements are then added to the webpage. Event handling is used to perform this operation when the user clicks the button.

Concepts Used

- JavaScript Classes and Objects
- Constructor
- DOM Selection using getElementById()
- Event Handling using addEventListener()
- Dynamic HTML Element Creation using createElement()
- Modifying Content using.textContent
- Adding Elements using appendChild()

Code

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Student Profile</title>
</head>
<body>
```

```
<h1>Student Details</h1>
```

```
<div class="container">
```

```
  <label for="name">Name: </label>
```

```
  <input type="text" placeholder="Enter your Name" id="name">
```

```
  <br><br>
```

```
  <label for="number">Roll Number: </label>
```

```
  <input type="text" placeholder="Enter your Roll Number"
id="number">
```

```
  <br><br>
```

```
  <label for="dept">Department: </label>
```

```
  <input type="text" placeholder="Enter your Department Name"
id="dept">
```

```
  <br><br>
```

```
  <label for="cgpa">CGPA: </label>
```

```
  <input type="text" placeholder="Enter your CGPA" id="cgpa">
```

```
  <br><br>
```

```
  <button id="createbtn">Create Profile</button>
```

```
  <div id="display-profile"></div>
```

```
</div>
```

```
<script src="./script.js"></script>
```

```
</body>
```

```
</html>
```

CSS Code:

```
body{
```

```
    background-color: beige;
```

```
    display: flex;
```

```
    flex-direction: column;
```

```
    margin: 0;
```

```
    justify-content: center;
```

```
    align-items: center;
```

```
    min-height: 100vh;
```

```
}
```

```
.container{
```

```
    background-color: aqua;
```

```
    display: flex;
```

```
    flex-direction: column;
```

```
    width: 550px;
```

```
    padding: 20px;
```

```
    box-shadow: 10px 20px 30px 40px rgba(0, 0, 0, 0.15);
```

```
    border-radius: 8px;
```

```
}
```

```
#name , #number, #dept, #cgpa{
```

```
    width: 500px;
```

```
    border-radius: 6px;
```

```
    border: none;
```

```
    background-color: azure;
```

```
}
```

```
button{
```

```
    width: 300px;
```

```
    margin-left: 200px;
```

```
    border-radius: 5px;
```

```
    border: none;
```

```
padding: 10px;
width: 200px;
color: blue;
}
button:hover {
    background-color: darkblue;
}

#display-profile {
    background-color: white;
    padding: 20px;
    margin-top: 20px;
    border-radius: 10px;
    text-align: left;
    box-shadow: 0px 2px 8px gray;
}

#display-profile h2 {
    text-align: center;
}
```

JavaScript Code

```
class Student {
    constructor(name, rollnumber, department, cgpa) {
        this.name = name;
        this.rollnumber = rollnumber;
        this.dept = department;
        this.cgpa = cgpa;
    }
}

const createBtn = document.getElementById("createbtn");
```

```
createBtn.addEventListener("click", function() {

    const name = document.getElementById("name").value;
    const rollnumber = document.getElementById("number").value;
    const department = document.getElementById("dept").value;
    const cgpa = document.getElementById("cgpa").value;

    const student = new Student(
        name,
        rollnumber,
        department,
        cgpa
    );

    const profile = document.getElementById("display-profile");

    profile.innerHTML = "";

    const heading = document.createElement("h2");
    heading.textContent = "Student Profile";

    const namePara = document.createElement("p");
    namePara.textContent = "Name: " + student.name;

    const rollPara = document.createElement("p");
    rollPara.textContent = "Roll Number: " + student.rollnumber;

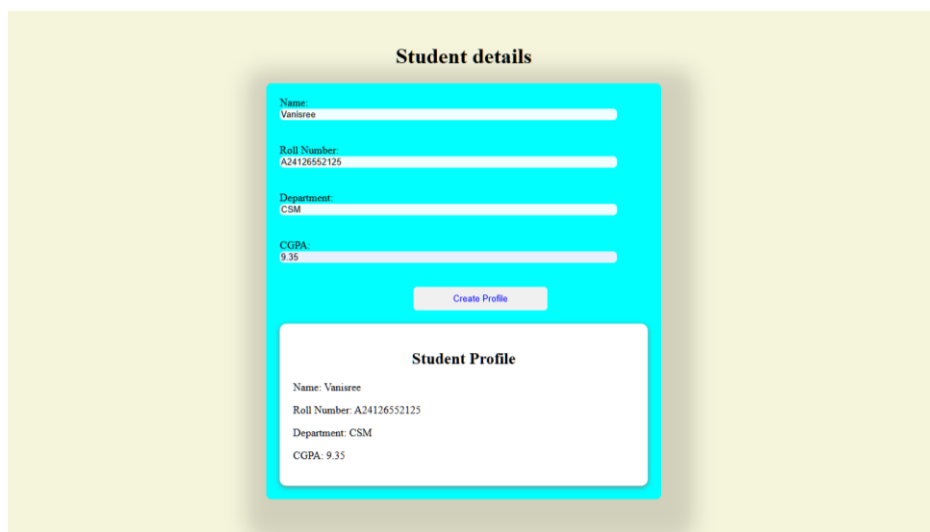
    const deptPara = document.createElement("p");
    deptPara.textContent = "Department: " + student.dept;

    const cgpaPara = document.createElement("p");
    cgpaPara.textContent = "CGPA: " + student.cgpa;
```

```
profile.appendChild(heading);  
profile.appendChild(namePara);  
profile.appendChild(rollPara);  
profile.appendChild(deptPara);  
profile.appendChild(cgpaPara);  
});
```

Output

When the user enters the student details and clicks the Create Profile button, the student profile is dynamically displayed on the webpage with the Name, Roll Number, Department, and CGPA.



The image shows a web application interface for creating a student profile. It features a form titled "Student details" with four input fields: "Name" (containing "Vanisree"), "Roll Number" (containing "A24126552125"), "Department" (containing "CSM"), and "CGPA" (containing "9.35"). A "Create Profile" button is located below the form. Below the button, a "Student Profile" section displays the entered information: "Name: Vanisree", "Roll Number: A24126552125", "Department: CSM", and "CGPA: 9.35".

Student details	
Name:	Vanisree
Roll Number:	A24126552125
Department:	CSM
CGPA:	9.35
<button>Create Profile</button>	
Student Profile	
Name:	Vanisree
Roll Number:	A24126552125
Department:	CSM
CGPA:	9.35